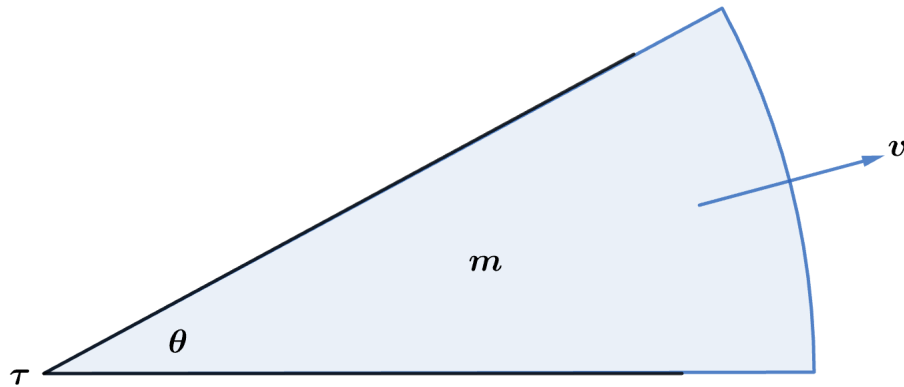


2019A F=ma Exam: Problem 18

Kevin S. Huang



The potential energy stored in the boards is

$$U = \tau\theta$$

since this is also the work done to open the boards to angle θ . This is transferred to the kinetic energy of the wedge as it leaves,

$$\frac{1}{2}mv^2 = \tau\theta$$

Thus,

$$v = \sqrt{\frac{2\tau\theta}{m}}$$

so the answer is D.